Slip 1 – IP & MAC Configuration

Q1. What is MAC address?

A MAC address is a unique physical address of a network device.

Q2. What is IPv4?

IPv4 is a 32-bit logical address used for communication.

Q3. What is subnet mask?

A subnet mask separates network and host portions.

Q4. What is default gateway?

A default gateway is a router address used to access other networks.

Q5. What is ping command?

Ping command checks connectivity between devices.

Q6. What is logical address?

A logical address is an IP address assigned to a device.

Q7. What is physical address?

A physical address is the hardware address of a device.

Q8. Difference between IP and MAC?

IP is a logical address while MAC is a physical address.

Slip 3 – Dotted Decimal to Binary

Q1. How many bits are in IPv4?

IPv4 contains 32 bits.

Q2. How many octets are in IPv4?

IPv4 contains four octets.

Q3. What is octet range?

Each octet ranges from 0 to 255.

Q4. What is binary notation?

Binary notation represents numbers using 0 and 1.

Q5. What is total IPv4 address space?

IPv4 has 2^{32} addresses.

Q6. What is dotted decimal notation?

Dotted decimal notation is decimal representation of an IP address.

Q7. Which IP is invalid?

An IP with octet greater than 255 is invalid.

Q8. What is IPv4 header?

IPv4 header contains packet information.

Slip 4 – Binary to Dotted Decimal

Q1. What is dotted decimal notation?

Dotted decimal notation represents IP in decimal form.

Q2. How many bits per octet?

Each octet contains 8 bits.

Q3. Which IP is invalid?

IP with octet value above 255 is invalid.

Q4. What is IPv4 header?

IPv4 header is the information section of packet.

Q5. Total address space of IPv4?

IPv4 has 4.3 billion addresses.

Q6. What is binary number system?

Binary system uses only 0 and 1 digits.

Q7. What is network address?

Network address identifies the network.

Q8. What is broadcast address?

Broadcast address sends data to all hosts.

Slip 5 – NetID and HostID

Q1. What is NetID?

NetID identifies the network.

Q2. What is HostID?

HostID identifies a device in a network.

Q3. What is /25?

/25 means 25 bits for network part.

Q4. What are usable hosts in /28?

A /28 network has 14 usable hosts.

Q5. Why use subnetting?

Subnetting divides networks efficiently.

Q6. What is subnet mask?

Subnet mask separates network and host parts.

Q7. What is network ID?

Network ID identifies the network address.

Q8. What is host address?

Host address identifies a specific device.

Slip 6 – IP Classes

Q1. How many IP classes exist?

There are five IP classes.

Q2. Class A range?

Class A range is 1–126.

Q3. Class B range?

Class B range is 128–191.

Q4. Class C range?

Class C range is 192–223.

Q5. Default mask of Class C?

Default mask of Class C is 255.255.255.0.

Q6. Which class supports largest hosts?

Class A supports largest hosts.

Q7. Which bits determine class?

Starting bits determine IP class.

Q8. Which class is used for multicast?

Class D is used for multicast.

Slip 7 – Arduino LED Pattern

Q1. What is Arduino?

Arduino is a microcontroller development board.

Q2. What is LED?

LED means Light Emitting Diode.

Q3. Function of pinMode()?

pinMode() sets pin as input or output.

Q4. Function of digitalWrite()?

digitalWrite() sets HIGH or LOW value.

Q5. What is Arduino IDE?

Arduino IDE is software for coding.

Q6. What is HIGH?

HIGH represents logic 1.

Q7. What is LOW?

LOW represents logic 0.

Q8. Why use LEDs?

LEDs are used for indication.

Slip 8 – Bluetooth

Q1. What is Bluetooth?

Bluetooth is wireless short-range communication technology.

Q2. Bluetooth range?

Bluetooth range is around 10 meters.

Q3. Bluetooth module used?

HC-05 or HC-06 module is used.

Q4. Bluetooth application?

Bluetooth is used for wireless data transfer.

Q5. Bluetooth frequency?

Bluetooth operates at 2.4 GHz.

Q6. Bluetooth works on which network?

Bluetooth works on PAN.

Q7. What is pairing?

Pairing connects Bluetooth devices.

Q8. What is Bluetooth used in IoT for?

Bluetooth is used for device communication.

Slip 10 – ZigBee Communication

Q1. What is ZigBee?

ZigBee is a low-power wireless protocol.

Q2. ZigBee range?

ZigBee range is 10–100 meters.

Q3. ZigBee application?

ZigBee is used in home automation.

Q4. Function of XBee?

XBee transmits wireless data.

Q5. What is point-to-point communication?

It is communication between two nodes.

Q6. ZigBee frequency?

ZigBee works at 2.4 GHz.

Q7. ZigBee advantage?

ZigBee consumes less power.

Q8. Which topology supports ZigBee?

Mesh topology supports ZigBee.

Slip 11 – Street Light Control

Q1. Which sensor is used?

LDR sensor is used.

Q2. LDR full form?

LDR means Light Dependent Resistor.

Q3. Function of Serial.begin(9600)?

It starts serial communication.

Q4. Why use smart street lights?

Smart street lights save energy.

Q5. Night condition in LDR?

Low light indicates night condition.

Q6. LDR resistance in dark?

LDR resistance is high in dark.

Q7. LDR resistance in light?

LDR resistance is low in light.

Q8. Main purpose of street light control?

It provides automatic switching.

Slip 12 – Environment Monitoring

Q1. Which sensor is used?

DHT11 sensor is used.

Q2. DHT11 measures?

DHT11 measures temperature and humidity.

Q3. Function of Serial.available()?

It checks incoming data.

Q4. Environment monitoring application?

It is used in weather monitoring.

Q5. Example sensors?

Temperature, humidity and gas sensors are used.

Q6. What is humidity?

Humidity is water vapor in air.

Q7. What is IoT?

IoT means Internet of Things.

Q8. Why monitor environment?

Environment monitoring tracks conditions.

Slip 13 – Security System

Q1. Which sensor is used?

PIR sensor is used.

Q2. PIR full form?

PIR means Passive Infrared Sensor.

Q3. Function of digitalWrite(9,HIGH)?

It sets pin output HIGH.

Q4. PIR detects?

PIR detects human motion.

Q5. Alarm use?

Alarm provides security alerts.

Q6. What does buzzer do?

Buzzer produces sound alerts.

Q7. What is intrusion?

Intrusion means unauthorized entry.

Q8. What is motion detection?

Motion detection senses movement.

Slip 14 – Door Control

Q1. Which motor is used?

Servo motor is used.

Q2. Function of servo motor?

Servo motor opens and closes doors.

Q3. Function of Serial.println()?

It prints output with a new line.

Q4. Which sensor is used?

Ultrasonic sensor is used.

Q5. Ultrasonic sensor measures?

Ultrasonic sensor measures distance.

Q6. What is trigger pin?

Trigger pin sends ultrasonic signals.

Q7. What is echo pin?

Echo pin receives reflected signals.

Q8. Purpose of smart door system?

Smart door system provides automatic access control.